

Owner Big Brain — Hackathon Briefing

Tuesday 29 September 2026 · Hemel office · 09:00 start · demo at 17:00

Read this before the day. Ten minutes.

✅ **Confirmed 24 September: we are building this natively in Snowflake.** Peter has tested the full chain end to end and it works. Gary has created the role. Everything you need is already enabled — see *What's already set up for you*.

Why we're doing this

Three separate requests about owners are open right now, across three business teams.

The ask	Who asked	Status today
Owner Lifetime Value — what an owner is worth, including costs	Rachel Gregory, Pitch Profitability Director	Requested, spec written
Top of Funnel — the prospect-to-owner journey	Caravan Sales + Data Engineering	In progress
Owner Attribution — which routes a lead actually takes	Marketing	Built, but standalone

Built separately, **these three will disagree** — different owner definitions, different cost bases, different attribution windows. We have just lived through exactly that on F&B revenue, where Data Science and reporting produced different numbers from one source and it took weeks to reconcile.

Built together they answer one question the business genuinely cannot answer today:

Which acquisition routes produce the owners worth having?

Not which produce the most leads. Which produce owners who stay, spend, and are worth the cost of acquiring.

The real problem, in one line from the LTV spec

"Top-of-funnel volume growth without an LTV understanding is value-destructive — if they only last a year or two years, we're making life harder for ourselves."

Owner acquisition is currently measured on **enquiry volume**, because that's the only number available. CPA is measured per *gross sign-up*, not per completion — so a channel with a 20% cancellation rate and one with 50% look identical. The only levers to convert that volume are free site fees or caravan price reductions, both of which destroy margin.

And the honest second reason

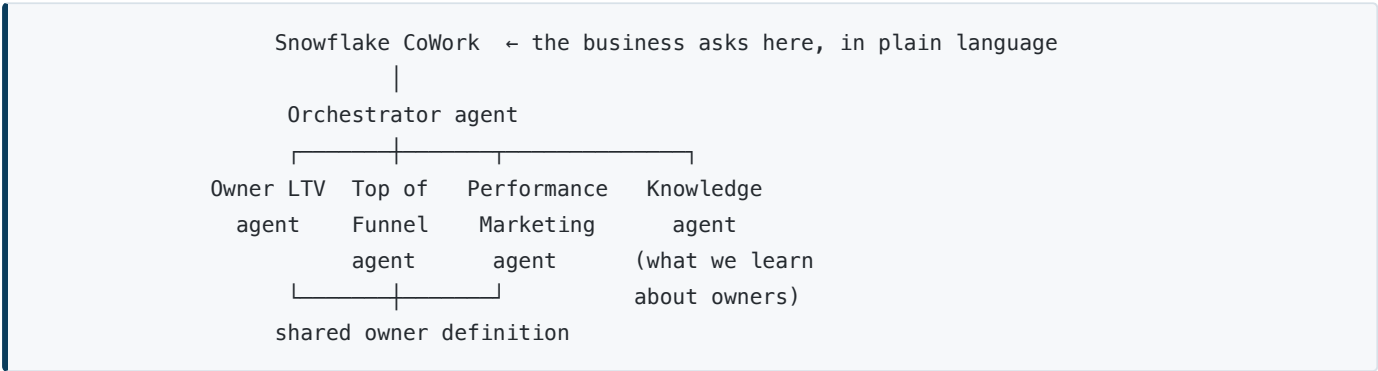
This team has never all been in one room. People in Budapest, Vilnius, Hemel and remote UK who have only ever met on a standup call. Data Engineering and Data Science have spent the year handing work over rather than building together.

Also: **until this week nobody on this team had built an agentic system or a semantic layer.** Peter has now — he built and tested the whole chain on 23–24 September — and he'll show you how at 09:25. Building that capability is a real objective of the day — not a side effect.

The team being in one room is most of the value. Nobody is judging this on how polished the models are.

What we're building

Three domain agents over a shared owner definition, all reachable from one prompt — plus a knowledge agent that remembers what gets asked and learned.



Each domain agent sits on a **semantic view** — a Snowflake object describing your tables in business language, so the agent can answer questions without anyone hand-writing SQL.

Three steps, and they map onto the day:

Step	What	When
1	Build the semantic layer — tables, joins, metrics, and the business descriptions	Morning
2	Stand up an agent on your semantic layer	Afternoon
3	Connect everything to an LLM via MCP so one prompt reaches all the agents	Platform team, all day

✅ **All Snowflake-native.** `CREATE SEMANTIC VIEW` , `CREATE AGENT` , MCP server, CoWork as the front end. No third-party tooling, no UI to build, no cross-vendor integration.

Your teams

Owner LTV	Top of Funnel	Performance Marketing	Platform
Rida <i>(lead)</i>	Matt <i>(lead)</i>	Dan C <i>(lead)</i>	Peter <i>(lead)</i>
Roland	Balint	Alina	Sarunas
Victor	Lewis	Judy	Gary
Abdul	Kirsty	Dan G	John P
Joe	Donovan	Jade	Charan
Elisha	Fraser	Ahsan	Ian

Your four leads have been briefed — Rida, Matt, Dan C and Peter. Go to them first on the day.

Each lead already knows their data model inside out: Matt built the top-of-funnel report *and* its Tableau front end; Dan C built the facts and dimensions in Performance Marketing plus the Sigma reporting; Rida and Roland have worked the owner data through Pitch Perfect, with Joe on the churn model that feeds it.

What each team owns

Team	Domain	Your main tables
Owner LTV	What an owner is worth and how long they stay. Owner account history + the churn model	OWNER_RISK_SCORES , OWNER_STATUS_TIME_SERIES_ANALYSIS , FCT_ACCOUNT_HISTORY , DIM_PARK
Top of Funnel	The prospect-to-owner journey. Enquiry → appointment → signup → completion	FCT_TOP_OF_THE_FUNNEL , DIM_TOP_OF_THE_FUNNEL , the BRIDGE_*_TO_CS_TIERED_XID family
Performance Marketing	Which channels and campaigns touch an owner on the way in. Bloomreach + attribution	FCT_ATTRIBUTION_JOURNEY_SUMMARY , FCT_ATTRIBUTION_CHANNEL_JOURNEY_ROLE , DIM_BLOOMREACH_CUSTOMER
Platform	Orchestration, MCP server, CoWork, the knowledge agent , and the demo	No semantic view of its own — owns integration

✅ The knowledge agent — new, and the most interesting thing we're building

The Platform team is also building a **knowledge agent**: as people ask the Big Brain questions, what it learns about owners gets **stored and reused**, rather than re-derived from scratch every time.

This is Peter's idea and it addresses a real problem. If every question is answered fresh, two people asking the same thing in slightly different ways get different answers. If validated findings persist — "*we know that owners acquired through this channel at this park last on average X*" — the system gets smarter and more consistent as it's used.

⚠️ **That's the difference between a query tool and something that accumulates corporate knowledge.** Worth watching even if you're not on Platform.

The questions the business will actually ask

⚠️ **Rachel Gregory is opening the day with these in person**, focusing on lifetime value. These come from the real requests — **pick the ones your domain owns and build to answer them.**

Rachel Gregory — Owner value and investment

- What is an owner worth over their lifetime, by park, pitch grade and van grade?
- Which owners are high-risk *and* high-value? (Today the churn programme treats all high-risk owners equally — it shouldn't.)
- What does an owner cost to acquire, against what they're worth?
- Which acquisition profiles systematically produce **short-tenure owners**?
- What's the LTV impact of a pitch move, a part exchange, or an owner taking up lettings?

Performance Marketing — Spend allocation

- Which channels bring owners with the highest lifetime value, **not just the most leads**?
- What's the LTV-to-CPA ratio by channel?
- Where should next quarter's budget go?
- Which campaigns touch owners who go on to stay five years or more?

Caravan Sales — Funnel efficiency

- Where in the funnel do prospects drop out, by park and by source?
- Which lead sources convert to a completion, not just to an appointment?

- How long does the journey from first enquiry to completion actually take?
- Which appointments show, and does the advisor make a difference?

The integration question — needs all three domains

"Show me acquisition channels ranked by the lifetime value of the owners they produced, against what we spent to get them."

⚠️ **No single domain can answer that.** It needs Performance Marketing's channel data, Top of Funnel's journey and cost, and Owner LTV's value — joined on a shared definition of an owner. **That's the demo.**

The day

Time	What	Who
09:00	Why we're here and what we're trying to do	Donovan
09:10	What the business is actually asking — customer lifetime value	Rachel Gregory
09:25	The tooling, and how it's set up — semantic views, agents, what Peter found building it	Peter
09:45	⚠️ Whole-team brainstorm: what is an owner, and how do we join this data? Common identifiers across all three domains	Everyone
10:30	Teams split. Step 1 — build your semantic layer	Domain teams
12:30	⚠️ Checkpoint — every team has a semantic view answering one real question. 🍕 Pizza. Working lunch	All
13:15	Step 2 — build your agent	Domain teams
14:30	⚠️ Checkpoint 2 — the orchestrator reaches all the agents	All
15:00	Verified queries for the demo questions	Domain teams
16:00	Rehearse. Fix what breaks	All
17:00	⚠️ Demo — the business comes back in	All
	Leave once we're done	

⚠️ Why 09:45 is the most important 45 minutes of the day

Two things get decided that cannot be parallelised: what counts as an owner, and how we join across the three domains.

On the definition: active only, or including leavers? The account or the person? Joint owners as one or two? If three teams each answer differently, we get three agents that confidently disagree — and the day reproduces the exact problem we're trying to solve.

On the joins: ⚠️ **this is the harder half.** Owner LTV keys on `ACCOUNT_NO` . Top of Funnel keys on `TOTF_TIERED_XID` , which itself resolves through email → Freshsales contact → analytics ID → plot owner. Performance Marketing keys on Bloomreach customer identity. **Getting from a marketing touch to an owner account is the single hardest technical problem of the day**, and it needs deciding by everyone, once, at the start.

Bring what you know about your domain's identifiers.

The 12:30 checkpoint — a view, not an agent

By pizza, every domain team should have a **working semantic view that answers one real question**. Not an agent. If you don't, say so at the checkpoint — that's what it's for, and there's still half a day.

⚠️ The demo has a benchmark

At 17:00 we're not just showing that it works. **We're taking the first question and asking it two ways:**

1. Through the framework we built — semantic layers, domain agents, orchestrator
2. Straight to Claude connected to Snowflake, with no semantic layer

Then we compare the answers, and how long each took.

⚠️ **This is a real test and it might not flatter us.** Claude-on-raw-Snowflake may be faster. It may even get the right answer. What we expect is that it's slower on complex questions, less consistent between runs, and unable to explain *which* definition it used — which is exactly the argument for a governed semantic layer.

✅ **If it turns out the simple approach is better, that's a genuinely valuable finding** and we report it. Nobody is being asked to make the framework look good.

✅ What's already set up for you

No setup on the day. Peter has tested this end to end and Gary has provisioned it.

Your role	✅ <code>NEXUS_SPIKE</code> — use this on the day. Everything you need is on it
What it can do	✅ Create semantic views, create agents, read the source data. All tested by Peter
Warehouse + database	✅ Dedicated to the day — Gary is confirming names by Monday 28th
Cortex	✅ Working, tested in our region

⚠️ **Check you can assume the role by Friday 25th.** Run:

```
USE ROLE NEXUS_SPIKE;
SELECT CURRENT_ROLE();
```

If that fails, tell Donovan or Gary on Friday 25th — not on the day. ⚠️ There is only Monday left after that.

⚠️ What you need to do before Tuesday

⚠️ **Two working days: Friday 25th and Monday 28th.**

Everyone

1. **Read this.** Nearly done.
1. ⚠️ **Check `NEXUS_SPIKE` works** — the two lines above.
1. **Look at a semantic view that already works**, on our own data: `` `sql DESCRIBE SEMANTIC VIEW NEXUS_PLATINUM.EPOS_SALES.SEM_EPOS_SALES;` `` ✅ **The best ten minutes of preparation available.** A real

semantic view over EPOS data, with real business context in it — GBP, rounding, how refunds work. You'll write something this shape.

- 1. **Skim the Snowflake docs** — [semantic views overview](#) and the [worked example](#). Twenty minutes.
- 1. ⚠️ **Read the validation rules** — [here](#). The two that will bite you: **semantic views reject many-to-many relationships, and reject joins where two tables can be reached by more than one path**. Knowing this before you model saves an afternoon.
- 1. **Come with your domain's identifiers in your head** for the 09:45 session. Which key does your data actually join on?

Your team — with your lead, before Tuesday

- 1. ⚠️ **Agree your 4–6 tables**. Not twenty. Semantic views get hard fast, and you have one day.
- 1. ⚠️ **Pre-flatten the problem tables into views**. Specifics below — each one is a rejection or a performance wall waiting to happen.
- 1. **Draft your five questions** — the ones your agent answers at 17:00. These become verified queries, the highest-value thing you'll do all day.

⚠️ **The specific traps, by team — verified against the real tables**

Team	Problem	Why it matters
Owner LTV	<code>FCT_ACCOUNT_HISTORY</code> is 350 million rows with only 7 columns	Filter to the latest snapshot in a view first, or every query crawls. ✅ The <code>current_pitch</code> CTE in <code>High_risk_owner_extract_v4.sql</code> already does this
Top of Funnel	<code>FCT_TOP_OF_THE_FUNNEL</code> has six date columns pointing at one calendar	That's a multi-path rejection . Pick the two or three the demo needs
Performance Marketing	<code>BRIDGE_BLOOMREACH_CUSTOMER_IDENTITY</code> is a 30 million-row bridge	That's the many-to-many semantic views reject. Flatten it
Performance Marketing	⚠️ Several <code>ATTRIBUTION_*</code> tables are empty while <code>FCT_ATTRIBUTION_*</code> have data	<code>ATTRIBUTION_CARAVAN_SALES_TOUCHPOINTS</code> = 0 rows ; <code>FCT_ATTRIBUTION_CARAVAN_SALES_PATH_TO_WEB_ENQUIRY</code> = 10M . You'd model an empty table and not find out until the demo. Start from <code>FCT_ATTRIBUTION_JOURNEY_SUMMARY</code>

✅ **Owner LTV — start here, not with table exploration**

`Product_Mgmt/product_scratchpad/High_risk_owner_extract_v4.sql` already contains your join path, your grain, **and the data caveats**. It documents that `ASSUMED_FINANCE_BALANCE` is a *proxy* for settlement because no real field exists, that `MONTHLY_INCOME_TOTAL` is a *proxy* for lettings cash, and that `PEX_BASE_TRADE_VALUE` is the right bottom-book-value column after Joseph's fix.

⚠️ **An agent that doesn't know those things will answer confidently and wrongly**. Put them in the semantic view's description and instructions — that's exactly what the field is for.

Pre-reading by team

Owner LTV

- `haven_data_science_artefacts/CLV/Owner_LTV_Development_Specification.md` — especially §4.1, the value components and our confidence in each
- `haven_data_science_artefacts/CLV/Owner LTV Brainstorm – Meeting Notes 20260731.md`

- `product_scratchpad/High_risk_owner_extract_v4.sql`

Top of Funnel

- `core-dbt#1426` and `#1427` — the gold fact and dimension models
- ⚠️ **The identity problem:** the model keys **89.3% of prospects on email address**, not Haven ID. Net-new and returning leads can't currently be separated, which inflates the CPA denominator. Appendix A of the LTV spec. **Bring this to the 09:45 session — it's directly relevant to the join discussion**

Performance Marketing

- `haven_data_artefacts/Performance_marketing/Attribution model for Haven Holidays.docx`
- `product_scratchpad/Attribution/`

Platform

- Snowflake `CREATE AGENT` — the agent specification reference, for how one agent references another
- Snowflake `Cortex Agents MCP server`
- Snowflake `CoWork`
- ✅ **Talk to Peter** — he has already built this chain and knows where the friction is
- `haven_data_science_artefacts/Planning/ai_enablement_strategy.md` §2.3 — read the next section first

⚠️ The problem we expect to hit — and want to

Peter raised this in the AI strategy session and it's the most important thing in this briefing.

He asked Claude to correlate a heatwave against F&B consumption using two parallel agents. **One decided a "warm year" meant the maximum temperature recorded. The other counted days above 20°C.** Same question, two entirely defensible methods, **different answers.**

"It's not really about data quality or what you show... It's derived procedures when the AI has multiple choices of getting really a more complicated result."

⚠️ **A semantic layer does not fix this.** Curating the data fixes *which numbers* an agent uses. It does not fix *which method* it picks. Three agents answering owner questions independently is precisely the setup that produces divergence.

Three consequences for the day:

1. **This is what verified queries are for.** Pin a known question to a known method and the agent has no choice to make. That's the 15:00 slot.
2. ✅ **This is what the knowledge agent is for.** If a validated finding persists, the next person asking gets the same answer rather than a freshly re-derived one.
3. ✅ **We should show it failing.** Ask the orchestrator something the owner definition doesn't pin down, and let it diverge in front of the business.

That last one is deliberate. Three agents agreeing by luck teaches nobody anything. ⚠️ **Overselling AI-on-data is how good ideas lose credibility** — and being straight about where it needs care is what earns the right to build more.

What we keep afterwards

	Why it lasts
✔ The agreed owner definition and join map	What three teams would otherwise each invent differently. The real deliverable
✔ Three semantic views with real business descriptions	Reusable by CoWork, Sigma, Tableau, or any future agent
✔ A team that has built an agentic system	Two weeks ago that was nobody. This is the capability gap closing
Three agents, an orchestrator, and a knowledge agent	The pattern, proven on our own data
⚠ The benchmark result	Framework vs raw Claude-on-Snowflake, honestly measured. Useful either way

What this day is not

- **Not** three finished models. One working thread.
- **Not** a delivery commitment. Nothing goes to production on Wednesday.
- **Not** a competition between teams. Integration is the point — three brilliant agents that can't talk to each other is a failed day.
- **Not** a test of you. If your domain turns out harder than expected, say so at the 12:30 checkpoint and narrow the scope. That's a finding, not a failure.

Questions

Ask your lead first — **Rida, Matt, Dan C or Peter**. Then Donovan.

⚠ **Tell someone by Friday 25th if:** `NEXUS_SPIKE` doesn't work for you, your tables look like they won't fit the constraints, or you think your domain is scoped wrong.

The day after: Snowflake World Tour at ExCeL, Wednesday 30th. Free, not mandatory. The Cortex and agentic tooling sessions are directly relevant to whatever we build on Tuesday — pick your sessions in advance, the agenda is large and it's easy to lose a day wandering.